Chimwemwe Chinkuyu
https://ckcgithub16.github.io/CC_Portfolio/

Education

Princeton University, Princeton, NJ — Class of 2026
B.S.E. Candidate, Mechanical and Aerospace Engineering
Relevant Coursework: Eng. Design, Mech. Design, Robotics and Intelligent Systems, Heat Transfer, Data Structures and Algorithms

Technical Skills

CAD/CAM Tools: PTC Creo, Fusion 360, Onshape, Autodesk Inventor
Software: Python, MATLAB, Java, C, Simulink, HTML
Fabrication and Analysis: FEA, 3D-printing (FDM, SLA, SLS), CNC milling, manual machining, lathing, laser cutting, basic GD&T
Robotics: Rhino and Grasshopper, ROS, UR3 Collaborative Robot programming, UR-RTDE

Internship Experience

Early Charm Ventures — Baltimore, MD
Engineering Intern — Jun. 2025-Aug. 2025
- Programmed and operated a MillRight CNC to machine 324 light fixture slats, developing Fusion 360 CAM toolpaths, reducing cycle time by up to 15%, and troubleshooted gantry and limit switch failures through manufacturer coordination.
- Supported R&D of high-temperature-resistant carbide ceramic 3D-printing by preparing powder blends, running Selective Laser Reaction Sintering (SLRS) experiments, and characterizing ceramic microstructures via SEM and XRD.
- Advanced development of an electromechanical patient privacy screen, including Fusion 360 CAD modeling, FDM 3D-printing of revised components, and troubleshooting an Arduino-based accelerometer alert circuit.

Ventura Medical Technologies — Barcelona, Spain
Engineering Intern — May 2023-Jul. 2023
- Enhanced Python/Jupyter tools to analyze DICOM files characterizing Pectus Excavatum, generating surgeon-specific implant placement reports; developed Multi-Planar Reconstruction (MPR) and multi-location sternal density scan modules.
- Analyzed 20+ CT scans in RadiAnt, producing patient-specific sternal measurement reports and surgical recommendations for Pectus Up implant placement.
- Researched and proposed redesign recommendations for the Lantitaps device prototype, including higher-voltage batteries and pumps with greater flow rate and pressure to meet target specifications.
- Supported regulatory and commercial operations by maintaining the Pectus Up patient database, coordinating with distributors for surgery data, and developing patient-facing brochure content for the Pectus Up surgical kit.

Early Charm Ventures — Baltimore, MD
Engineering Intern — Jun. 2022-Aug. 2022
- Ran and troubleshooted Prusa Mini +, Formlabs 3, 3L, and Fuse 1 3D printers and failed prints.
- Collaborated with engineers and scientists to analyze and optimize customer .stl files for printability.
- Postprocessed hundreds of FDM, FFF, SLA, and SLS parts in numerous materials to customer satisfaction.
- Contributed to the optimization of fast-paced 3D-printing startup workflow as it increased production volume.

Projects

Design and Manufacturing of a Search and Rescue Robot — Oct. 2025-Dec. 2025
- Collaborated with 10 classmates to design and build a 120-lb marsupial search-and-rescue robot, implementing closed-loop navigation via light and ultrasonic sensing for successful autonomous terrain traversal and sub-robot deployment.
- Led design, FEA, and manufacturing of low-carbon steel chassis and 60:1 belt-driven drivetrain.

Autonomous RC Race Truck — Mar. 2025-May 2025
- Collaborated with 3 classmates to program and tune an iterative Linear Quadratic Regulator (iLQR) algorithm in Python using ROS to autonomously navigate an RC truck through an obstacle course, placing 4th fastest out of 14 teams.
- Developed a real-time trajectory planning system with dynamic lane-switching to reach 16 waypoints while avoiding obstacles.
- Designed a safety filter to evaluate manual driving inputs and override them only when predicted to cause unsafe lane departures, balancing driver autonomy with collision prevention.

Activities

Princeton Christian Fellowship, Ministry Team Leader — Jan. 2023-Present
Princeton University Robotics Club — Princeton, NJ
Member of the Mechanical Team — Sep. 2022-Present
- Designing and building a Myoelectric Prosthetic Hand using Onshape and 3D-printing.
- Designed and built a BB-8 Droid from *Star Wars.*